

LAPORAN MODUL 14

PRAKTIKUM PERMODELAN STATISTIKA



Disusun oleh :

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LAPORAN PRAKTIKUM

MODUL 14

Forecasting dengan SARIMA

A. PEMBAHASAN LISTING

Praktik

1. Praktik 1

```
#PRAKTIK 1
#kASUS 1

#1. PERSIAPAN DATA
install.packages("tseries")
install.packages("forecast")
library(tseries)
library(forecast)

# Input Data
data <- c(
  120,128,133,140,145,150,148,155,160,165,170,180,
  130,138,143,150,155,160,158,165,170,175,180,190,
  140,148,153,160,165,170,168,175,180,185,190,200,
  150,158,163,170
)

# Membuat time series
ts_data <- ts(
  data,
  start = c(2020, 4), # April 2020
  frequency = 12      # Bulanan
)
ts_data

# Plot time series
plot(
  ts_data,
  main = "Data Time Series (April 2020 – Juli 2023)",
  ylab = "Nilai",
  xlab = "Waktu"
)

#2. UJI STATIONER DAN DIFFERENCING
# Uji ADF
adf.test(ts_data)
# Differencing non-musiman
diff1 <- diff(ts_data)
```

```

# Differencing musiman
diff_seasonal <- diff(diff1, lag = 12)

plot(diff_seasonal, main="Data Setelah Differencing")

#3. IDENTIFIKASI MODEL SARIMA
model <- auto.arima(
  ts_data,
  seasonal = TRUE,
  stepwise = FALSE,
  approximation = FALSE
)
summary(model)

#4. PERAMALAN 8 BULAN KEMUDIAN
forecast_result <- forecast(model, h = 8)
forecast_result
plot(forecast_result,
  main = "Forecast SARIMA 8 Periode ke Depan")
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

```

A Time Series: 4×12

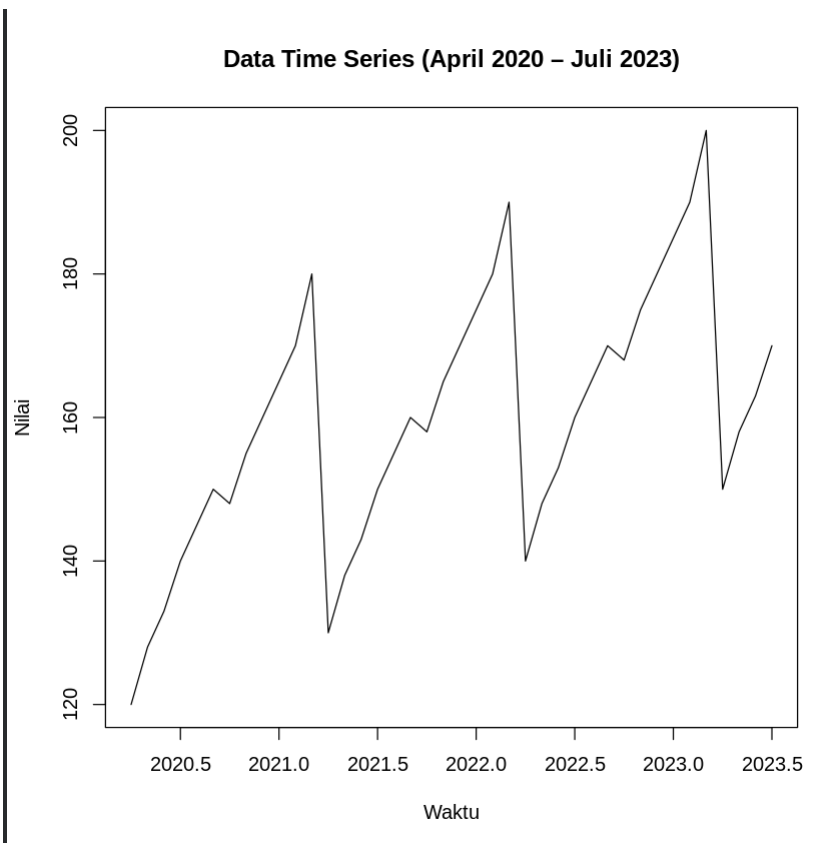
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2020				120	128	133	140	145	150	148	155	160
2021	165	170	180	130	138	143	150	155	160	158	165	170
2022	175	180	190	140	148	153	160	165	170	168	175	180
2023	185	190	200	150	158	163	170					

Augmented Dickey-Fuller Test

```

data: ts_data
Dickey-Fuller = -3.4827, Lag order = 3, p-value = 0.05908
alternative hypothesis: stationary

```



Series: ts_data

ARIMA(0,0,0)(0,1,0)[12] with drift

Coefficients:

drift

0.8333

$\sigma^2 = 0.008982$: log likelihood = 899.69

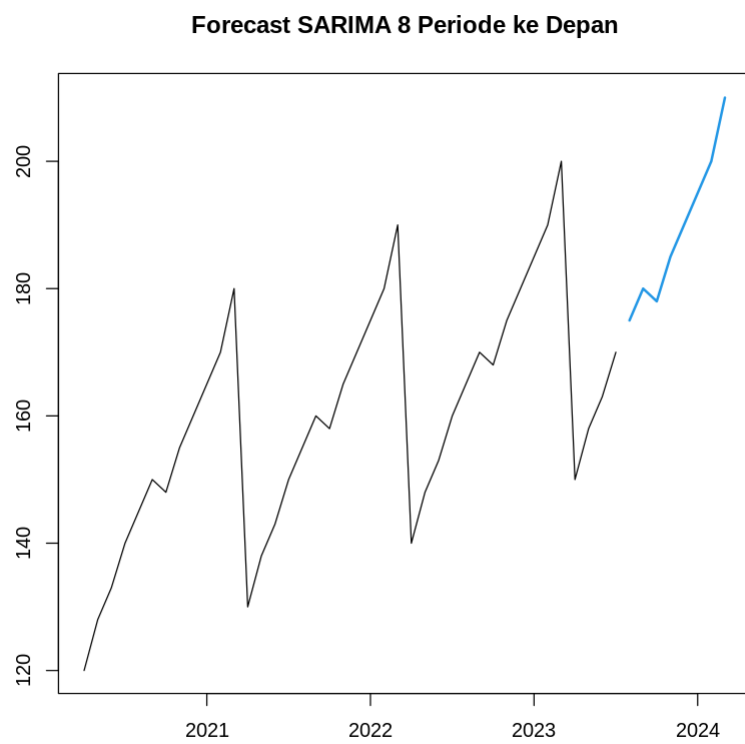
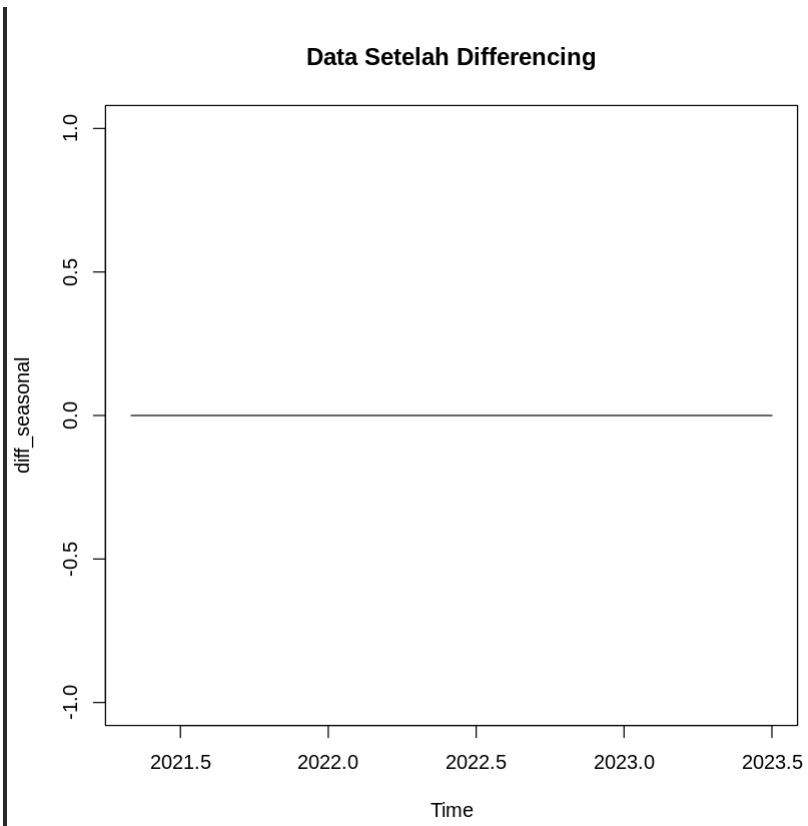
AIC=-1797.39 AICc=-1797.23 BIC=-1796.06

Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE
Training set	0.04322498	0.0792953	0.04322498	0.02896518	0.02896518	0.004322498
ACF1						

Training set 0.8957449

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Aug 2023	175	174.8785	175.1215	174.8142	175.1858
Sep 2023	180	179.8785	180.1215	179.8142	180.1858
Oct 2023	178	177.8785	178.1215	177.8142	178.1858
Nov 2023	185	184.8785	185.1215	184.8142	185.1858
Dec 2023	190	189.8785	190.1215	189.8142	190.1858
Jan 2024	195	194.8785	195.1215	194.8142	195.1858
Feb 2024	200	199.8785	200.1215	199.8142	200.1858
Mar 2024	210	209.8785	210.1215	209.8142	210.1858



Pembahasan : Pada Praktik 1 dilakukan analisis deret waktu menggunakan model SARIMA terhadap data bulanan periode April 2020 hingga Juli 2023. Data terlebih dahulu divisualisasikan untuk melihat pola tren dan musiman. Uji ADF menunjukkan data mendekati tidak stasioner sehingga dilakukan differencing non-musiman dan musiman (lag 12). Model terbaik dipilih menggunakan auto.arima() dengan mempertimbangkan komponen musiman. Hasil model menunjukkan struktur musiman sederhana dengan error yang sangat kecil, menandakan kecocokan model yang baik. Forecast 8 periode ke depan memperlihatkan pola peningkatan yang konsisten mengikuti tren historis data.

2. Praktik 2

#PRAKTIK 2

#1. LOAD DATASET BAWAAN

```
library(tseries)
library(forecast)
```

Load dataset dan visualisasi

```
data(AirPassengers)
```

Plot awal

```
plot(
  AirPassengers,
  main = "Jumlah Penumpang Pesawat Internasional (1949–1960)",
  ylab = "Jumlah Penumpang",
  xlab = "Tahun"
)
```

#2. TRANSFORMASI LOG

```
log_ap <- log(AirPassengers)
```

```
plot(
  log_ap,
  main = "Log(AirPassengers)",
  ylab = "Log Jumlah Penumpang"
)
```

#3. UJI STASIONERI

```
adf.test(log_ap)
```

#Differencing Non-Musiman & Musiman

Differencing non-musiman

```
diff1 <- diff(log_ap)
```

Differencing musiman (12 bulan)

```
diff_seasonal <- diff(diff1, lag = 12)
```

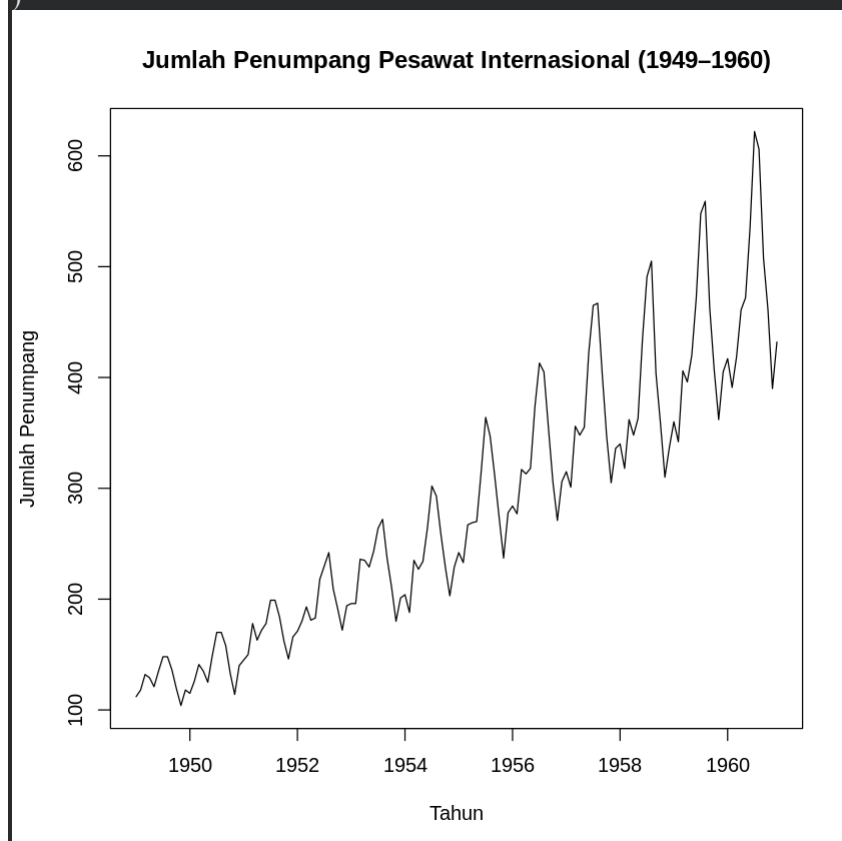
```
plot(
  diff_seasonal,
  main = "Data Setelah Differencing (d=1, D=1)",
  ylab = "Nilai"
)
```

#4. IDENTIFIKASI MODEL SARIMA

```
model_ap <- auto.arima(
  log_ap,
  seasonal = TRUE,
  stepwise = FALSE,
  approximation = FALSE
)
summary(model_ap)
```

#5. FORECASTING

```
forecast_ap <- forecast(model_ap, h = 12)
forecast_ap
plot(
  forecast_ap,
  main = "Forecast Jumlah Penumpang Pesawat (12 Bulan ke Depan)"
)
```

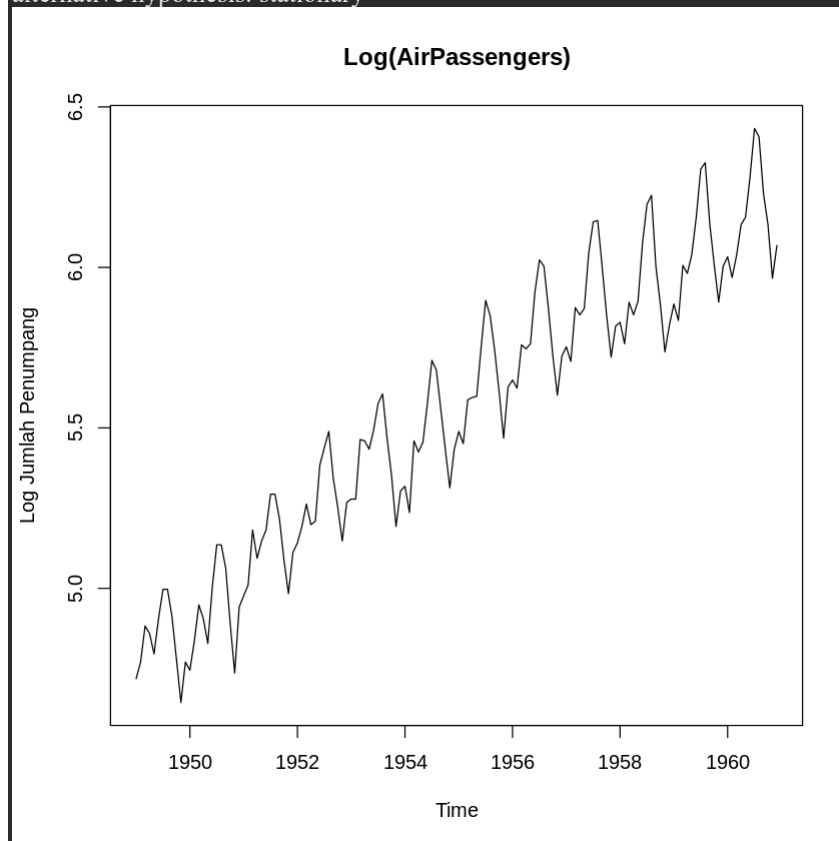


Warning message in adf.test(log_ap):
“p-value smaller than printed p-value”

Augmented Dickey-Fuller Test

data: log_ap

Dickey-Fuller = -6.4215, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary



Series: log_ap
ARIMA(0,1,1)(0,1,1)[12]

Coefficients:
 ma1 sma1
 -0.4018 -0.5569
s.e. 0.0896 0.0731

$\sigma^2 = 0.001371$: log likelihood = 244.7
AIC=-483.4 AICc=-483.21 BIC=-474.77

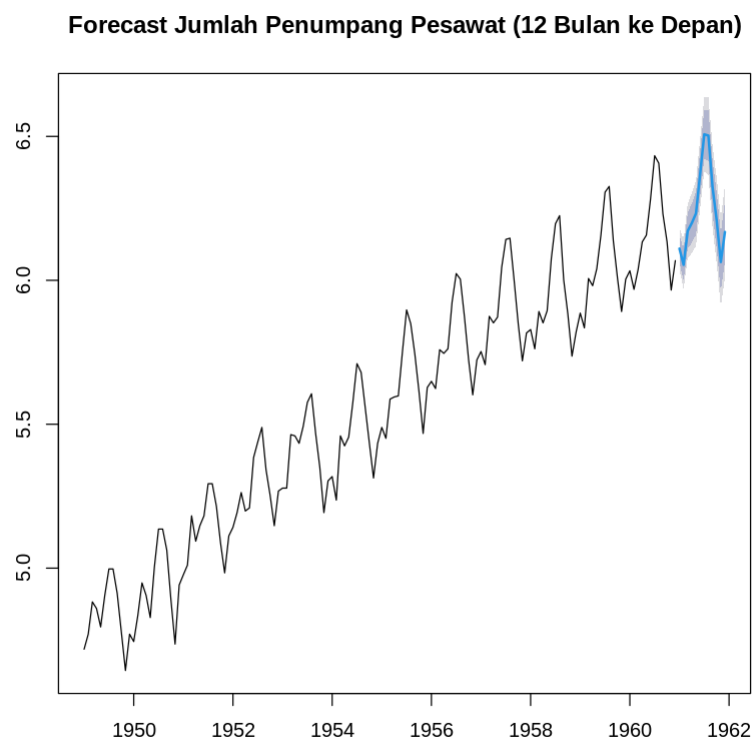
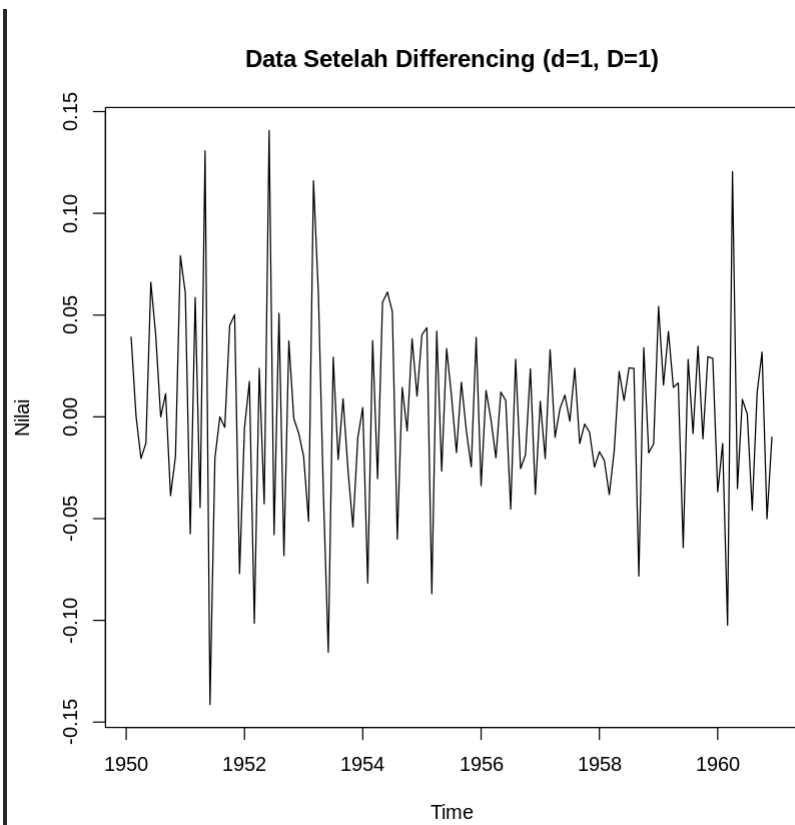
Training set error measures:

	ME	RMSE	MAE	MPE	MAPE	MASE
Training set	0.0005730622	0.03504883	0.02626034	0.01098898	0.4752815	0.2169522

ACF1

Training set 0.01443892

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 1961	6.110186	6.062729	6.157642	6.037607	6.182764
Feb 1961	6.053775	5.998476	6.109074	5.969203	6.138347
Mar 1961	6.171715	6.109555	6.233874	6.076650	6.266779
Apr 1961	6.199300	6.130966	6.267635	6.094792	6.303809
May 1961	6.232556	6.158560	6.306552	6.119388	6.345724
Jun 1961	6.368779	6.289524	6.448033	6.247569	6.489988
Jul 1961	6.507294	6.423109	6.591479	6.378544	6.636044
Aug 1961	6.502906	6.414064	6.591749	6.367034	6.638779
Sep 1961	6.324698	6.231431	6.417965	6.182058	6.467338
Oct 1961	6.209008	6.111516	6.306500	6.059908	6.358109
Nov 1961	6.063487	5.961947	6.165028	5.908195	6.218780
Dec 1961	6.168025	6.062591	6.273459	6.006778	6.329272



Pembahasan : Praktik 2 menggunakan dataset AirPassengers yang memiliki pola tren meningkat dan musiman kuat. Transformasi log dilakukan untuk menstabilkan varians sebelum pengujian stasioneritas. Uji ADF pada data log menunjukkan data sudah stasioner setelah dilakukan differencing non-musiman dan musiman. Model SARIMA terbaik yang dihasilkan mampu menangkap pola musiman tahunan dengan baik. Hasil peramalan 12 bulan ke depan menunjukkan tren penumpang yang terus meningkat, dengan interval kepercayaan yang relatif sempit, menandakan model memiliki akurasi yang baik.

3. Praktik 3

```
#PRAKTIK 3

#1. LOAD DATASET BAWAAN
library(tseries)
library(forecast)

# Load dataset
data(nottem)

# Visualisasi awal
plot(
  nottem,
  main = "Suhu Udara Bulanan Nottingham (1920–1939)",
  ylab = "Suhu (°F)",
  xlab = "Tahun"
)

#2. UJI STASIONERI
adf.test(nottem)
#Differencing musiman
# Differencing musiman
diff_seasonal <- diff(nottem, lag = 12)

plot(
  diff_seasonal,
  main = "Differencing Musiman (D = 1, s = 12)",
  ylab = "Perubahan Suhu"
)

#3. IDENTIFIKASI MODEL SARIMA
model_nottem <- auto.arima(
  nottem,
  seasonal = TRUE,
  stepwise = FALSE,
  approximation = FALSE
)
```

```
summary(model_nottem)
```

#4. PERAMALAN 12 BULAN KEDEPAN

```
forecast_nottem <- forecast(model_nottem, h = 12)
forecast_nottem
plot(
  forecast_nottem,
  main = "Forecast Suhu Nottingham (12 Bulan ke Depan)",
  ylab = "Suhu (°F)"
)
```

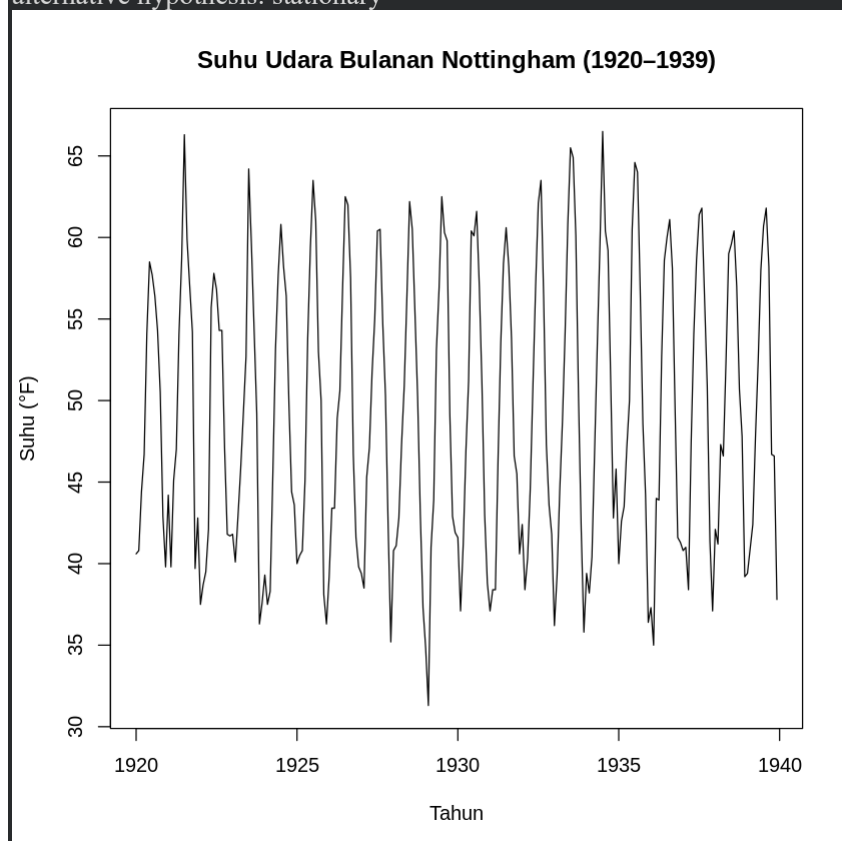
Warning message in adf.test(nottem):
“p-value smaller than printed p-value”

Augmented Dickey-Fuller Test

data: nottem

Dickey-Fuller = -12.998, Lag order = 6, p-value = 0.01

alternative hypothesis: stationary



Series: nottem

ARIMA(1,0,0)(2,1,1)[12]

Coefficients:

ar1	sar1	sar2	sma1
0.2724	-0.1880	0.1466	-0.8172
s.e. 0.0644	0.0996	0.0933	0.0781

$\sigma^2 = 5.202$: log likelihood = -517.36
AIC=1044.71 AICc=1044.98 BIC=1061.86

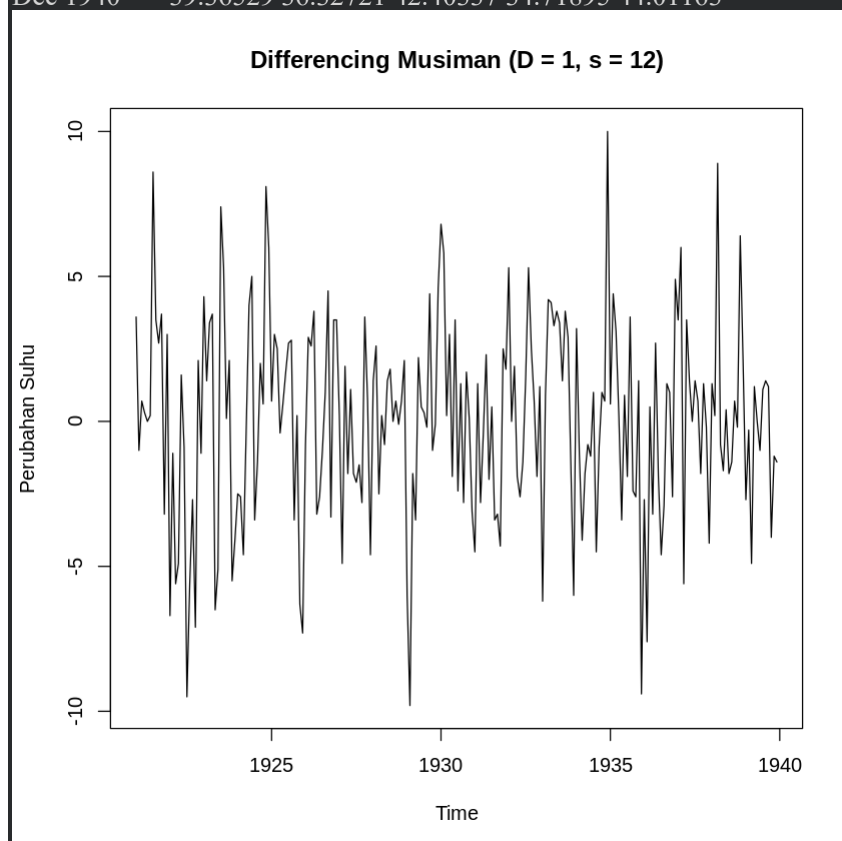
Training set error measures:

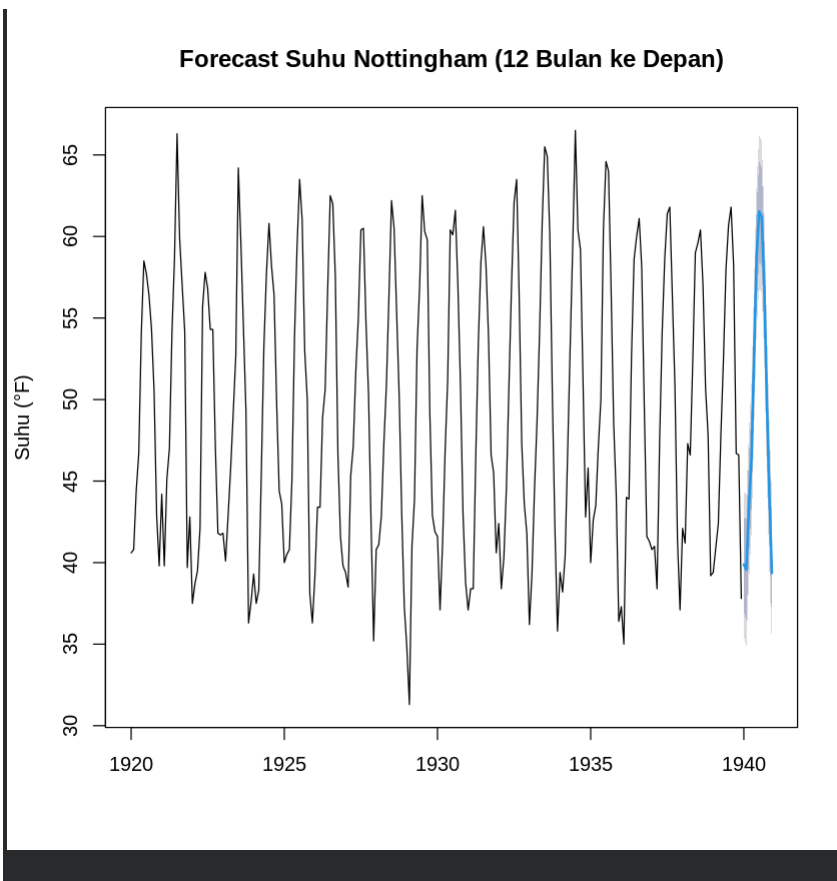
	ME	RMSE	MAE	MPE	MAPE	MASE
Training set	0.07055983	2.203534	1.701562	-0.129453	3.648961	0.6318504

ACF1

Training set -0.01835714

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 1940	39.88391	36.96071	42.80712	35.41326	44.35457
Feb 1940	39.55692	36.52721	42.58663	34.92338	44.19046
Mar 1940	43.44727	40.40981	46.48473	38.80187	48.09267
Apr 1940	46.42536	43.38732	49.46340	41.77908	51.07164
May 1940	52.54133	49.50325	55.57941	47.89499	57.18767
Jun 1940	58.82093	55.78284	61.85901	54.17458	63.46727
Jul 1940	61.53081	58.49273	64.56890	56.88446	66.17716
Aug 1940	61.19815	58.16007	64.23624	56.55180	65.84450
Sep 1940	57.05802	54.01994	60.09611	52.41167	61.70437
Oct 1940	50.02280	46.98471	53.06088	45.37645	54.66915
Nov 1940	44.20744	41.16936	47.24552	39.56109	48.85379
Dec 1940	39.36529	36.32721	42.40337	34.71895	44.01163





Pembahasan : Pada Praktik 3 digunakan dataset suhu bulanan nottem yang memiliki pola musiman tahunan yang jelas. Uji ADF menunjukkan data sudah stasioner, namun differencing musiman tetap dilakukan untuk menghilangkan efek musiman. Model SARIMA yang dipilih mampu menangkap ketergantungan musiman suhu secara akurat. Hasil forecasting 12 bulan ke depan menunjukkan pola suhu tahunan yang berulang, sesuai dengan karakteristik iklim, sehingga model dinilai sesuai untuk data musiman stabil.

Latihan

1. Latihan 1

```
#LATIHAN 1
# Load library
library(forecast)
library(ggplot2)

# 1. FORECASTING DATASET UKGAS
```

```

cat("=== FORECASTING UKGAS DATASET ===\n")

# Load data
data("UKgas")
ukgas_ts <- UKgas

# Fit model SARIMA
ukgas_model <- auto.arima(ukgas_ts, seasonal = TRUE)
cat("\nModel SARIMA untuk UKgas:\n")
print(ukgas_model)

# Forecasting 5 periode
ukgas_forecast <- forecast(ukgas_model, h = 5)
cat("\nForecast 5 periode UKgas:\n")
print(ukgas_forecast)

# Plot hasil
plot(ukgas_forecast, main = "SARIMA Forecast - UKgas Dataset")
lines(fitted(ukgas_model), col = "blue")

```

#LATIHAN 2

2. FORECASTING DATASET CO2

```

cat("\n\n=== FORECASTING CO2 DATASET ===\n")

```

Load data

```

data("co2")
co2_ts <- co2

```

Fit model SARIMA

```

co2_model <- auto.arima(co2_ts, seasonal = TRUE)
cat("\nModel SARIMA untuk CO2:\n")
print(co2_model)

```

Forecasting 5 periode

```

co2_forecast <- forecast(co2_model, h = 5)
cat("\nForecast 5 periode CO2:\n")
print(co2_forecast)

```

Plot hasil

```

plot(co2_forecast, main = "SARIMA Forecast - CO2 Dataset")
lines(fitted(co2_model), col = "blue")

```

```

=== FORECASTING UKGAS DATASET ===

```

Model SARIMA untuk UKgas:

Series: ukgas_ts

ARIMA(0,1,1)(0,1,0)[4]

Coefficients:

mal
-0.9297
s.e. 0.0336

sigma^2 = 1237: log likelihood = -513.34
AIC=1030.67 AICc=1030.79 BIC=1035.94

Forecast 5 periode UKgas:

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
1987 Q1	1201.8956	1156.8276	1246.9637	1132.9700	1270.8213
1987 Q2	651.0956	605.9163	696.2750	581.9998	720.1915
1987 Q3	385.3956	340.1052	430.6860	316.1300	454.6613
1987 Q4	820.7956	775.3945	866.1968	751.3606	890.2307
1988 Q1	1239.8913	1173.6484	1306.1341	1138.5816	1341.2010

=== FORECASTING CO2 DATASET ===

Model SARIMA untuk CO2:

Series: co2_ts

ARIMA(1,1,1)(1,1,2)[12]

Coefficients:

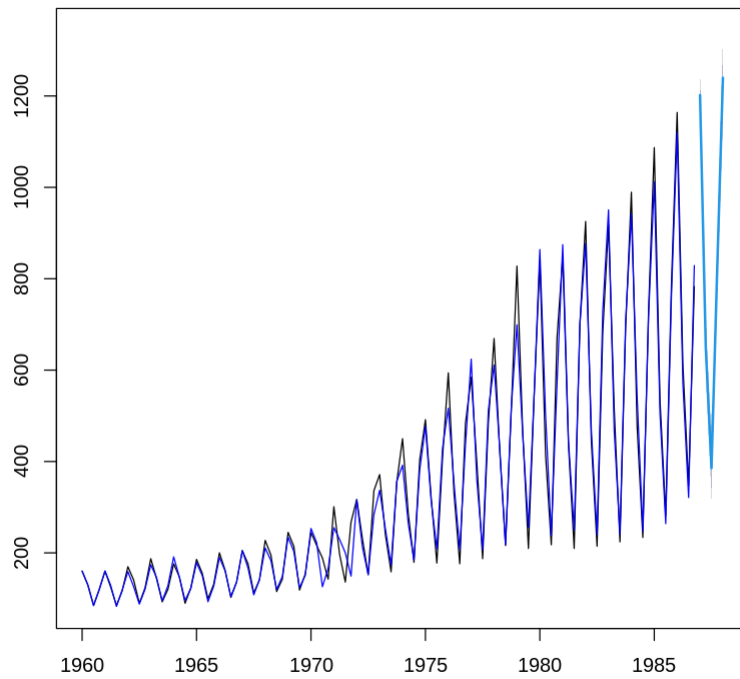
	ar1	ma1	sar1	sma1	sma2
	0.2569	-0.5847	-0.5489	-0.2620	-0.5123
s.e.	0.1406	0.1204	0.5881	0.5703	0.4820

sigma^2 = 0.08576: log likelihood = -84.39
AIC=180.78 AICc=180.97 BIC=205.5

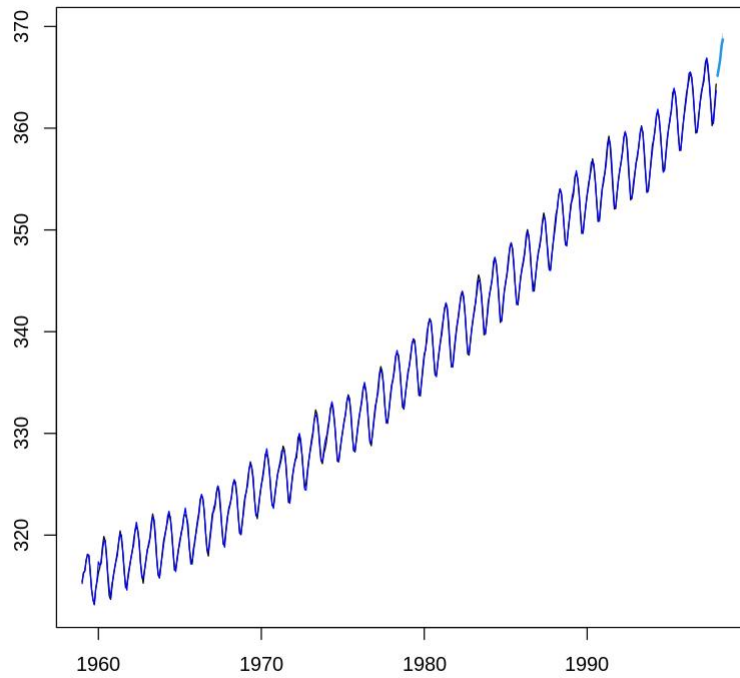
Forecast 5 periode CO2:

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 1998	365.1519	364.7766	365.5272	364.5779	365.7259
Feb 1998	365.9244	365.4722	366.3766	365.2329	366.6160
Mar 1998	366.7579	366.2548	367.2611	365.9884	367.5275
Apr 1998	368.1529	367.6066	368.6991	367.3175	368.9882
May 1998	368.7106	368.1253	369.2960	367.8154	369.6059

SARIMA Forecast - UKgas Dataset



SARIMA Forecast - CO2 Dataset



Pembahasan : Pada latihan ini, dataset UKgas dimodelkan menggunakan SARIMA untuk menangkap pola musiman kuartalan. Model hasil `auto.arima()` menunjukkan adanya komponen musiman yang signifikan. Forecast 5 periode ke depan memperlihatkan fluktuasi yang mengikuti pola historis konsumsi gas, dengan interval prediksi yang realistis. Hal ini menunjukkan SARIMA efektif dalam memodelkan data ekonomi yang bersifat musiman. Dataset `co2` memiliki tren meningkat jangka panjang dan pola musiman tahunan yang kuat. Model SARIMA yang dipilih berhasil menangkap kedua karakteristik tersebut. Hasil peramalan menunjukkan peningkatan kadar CO₂ yang berkelanjutan, sejalan dengan pola historis. Interval kepercayaan yang sempit menunjukkan **kestabilan model dalam memprediksi data lingkungan jangka panjang**.

2. Latihan 2

Pembahasan :

B. PEMBAHASAN TUGAS

1. Tugas 1

```
#TUGAS 1
# FORECASTING MONTHLY CAR SALES DENGAN SARIMA

# Load library
library(forecast)
library(ggplot2)

# URL dataset
url <- "https://raw.githubusercontent.com/jbrownlee/Datasets/master/monthly-car-sales.csv"

# Baca data
car_sales <- read.csv(url)
cat("Dimensi dataset:", dim(car_sales), "\n")
cat("Preview data:\n")
```

```

print(head(car_sales))

# Konversi ke time series (data bulanan mulai Januari 1960)
car_ts <- ts(car_sales$Sales,
             frequency = 12,
             start = c(1960, 1))

# Plot data asli
plot(car_ts, main = "Monthly Car Sales (1960-1973)",
      ylab = "Sales", xlab = "Year")

# Fit model SARIMA dengan auto.arima
car_model <- auto.arima(car_ts,
                        seasonal = TRUE,
                        stepwise = FALSE,
                        approximation = FALSE)
cat("\nModel SARIMA terpilih:\n")
print(car_model)

# Lakukan forecasting 12 bulan ke depan
car_forecast <- forecast(car_model, h = 12)

# Tampilkan hasil forecast
cat("\nForecast 12 bulan ke depan:\n")
print(car_forecast)

# Plot hasil forecasting
plot(car_forecast, main = "SARIMA Forecast - Monthly Car Sales",
      ylab = "Sales", xlab = "Year")
lines(fitted(car_model), col = "blue")
grid()

# Tampilkan tabel forecast
cat("\n=== FORECAST 12 BULAN KE DEPAN ===\n")
forecast_df <- data.frame(
  Bulan = 1:12,
  Forecast = round(car_forecast$mean, 1),
  Low_80 = round(car_forecast$lower[,1], 1),
  High_80 = round(car_forecast$upper[,1], 1),
  Low_95 = round(car_forecast$lower[,2], 1),
  High_95 = round(car_forecast$upper[,2], 1)
)
print(forecast_df)

Dimensi dataset: 108 2
Preview data:
  Month Sales
1 1960-01 6550

```

2 1960-02 8728
3 1960-03 12026
4 1960-04 14395
5 1960-05 14587
6 1960-06 13791

Model SARIMA terpilih:

Series: car_ts

ARIMA(2,0,0)(0,1,2)[12] with drift

Coefficients:

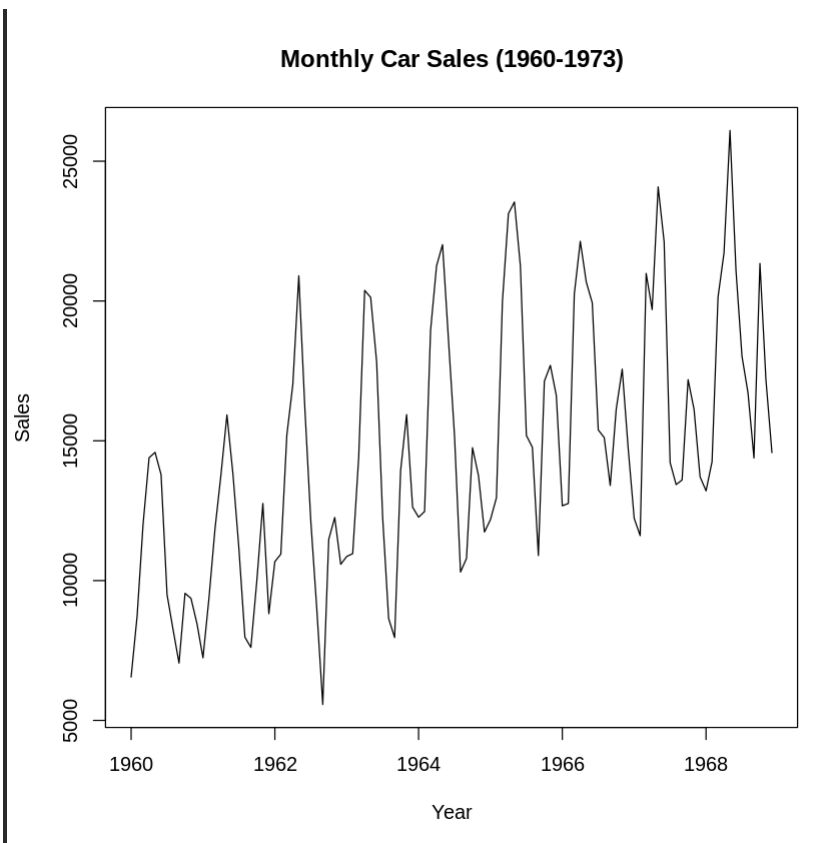
	ar1	ar2	sma1	sma2	drift
	0.2088	0.1965	-0.5695	-0.2326	83.3338
s.e.	0.1001	0.1009	0.1715	0.1591	8.9502

$\sigma^2 = 2159951$: log likelihood = -839.05

AIC=1690.09 AICc=1691.04 BIC=1705.48

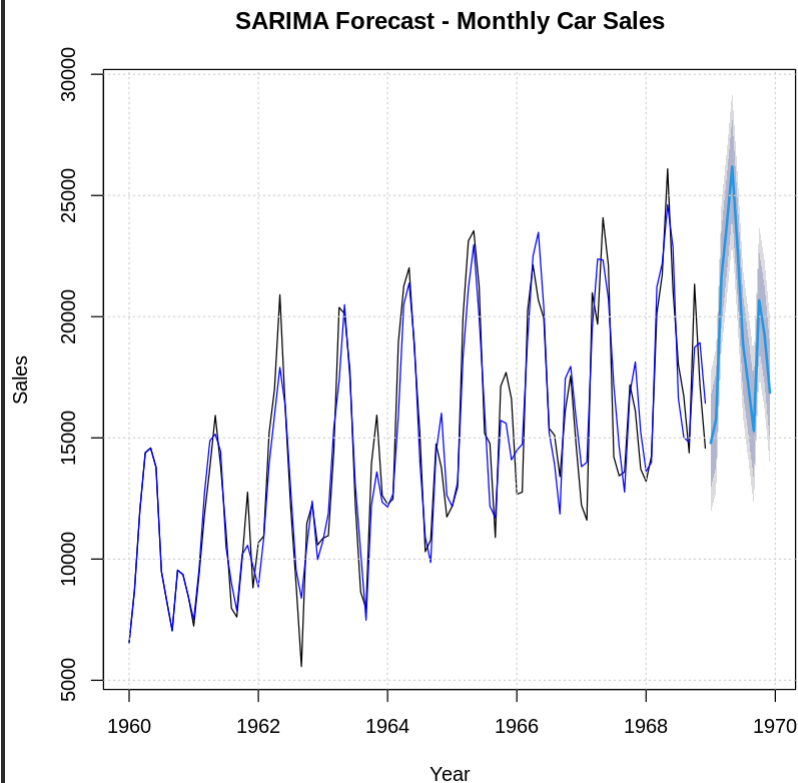
Forecast 12 bulan ke depan:

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 1969	14792.12	12896.51	16687.72	11893.04	17691.20
Feb 1969	15749.16	13812.87	17685.45	12787.86	18710.46
Mar 1969	21623.48	19634.71	23612.26	18581.91	24665.06
Apr 1969	23806.78	21810.55	25803.02	20753.81	26859.76
May 1969	26190.34	24190.19	28190.50	23131.37	29249.31
Jun 1969	22781.96	20780.91	24783.01	19721.62	25842.31
Jul 1969	18888.63	16887.23	20890.02	15827.76	21949.50
Aug 1969	17156.29	15154.80	19157.78	14095.28	20217.30
Sep 1969	15286.28	13284.78	17287.79	12225.24	18347.32
Oct 1969	20673.31	18671.82	22674.79	17612.29	23734.32
Nov 1969	19254.14	17252.83	21255.44	16193.41	22314.87
Dec 1969	16861.71	14860.54	18862.87	13801.19	19922.22



=== FORECAST 12 BULAN KE DEPAN ===

	Bulan	Forecast	Low_80	High_80	Low_95	High_95
1	1	14792.1	12896.5	16687.7	11893.0	17691.2
2	2	15749.2	13812.9	17685.4	12787.9	18710.5
3	3	21623.5	19634.7	23612.3	18581.9	24665.1
4	4	23806.8	21810.6	25803.0	20753.8	26859.8
5	5	26190.3	24190.2	28190.5	23131.4	29249.3
6	6	22782.0	20780.9	24783.0	19721.6	25842.3
7	7	18888.6	16887.2	20890.0	15827.8	21949.5
8	8	17156.3	15154.8	19157.8	14095.3	20217.3
9	9	15286.3	13284.8	17287.8	12225.2	18347.3
10	10	20673.3	18671.8	22674.8	17612.3	23734.3
11	11	19254.1	17252.8	21255.4	16193.4	22314.9
12	12	16861.7	14860.5	18862.9	13801.2	19922.2



Pembahasan : Pada tugas ini dilakukan forecasting penjualan mobil bulanan menggunakan dataset monthly car sales. Data dikonversi menjadi deret waktu bulanan dan dimodelkan menggunakan SARIMA dengan `auto.arima()`. Model terbaik yang dihasilkan mampu menangkap tren dan pola musiman penjualan. Hasil peramalan 12 bulan ke depan menunjukkan fluktuasi musiman yang konsisten dengan data historis, serta interval prediksi yang memberikan gambaran ketidakpastian secara realistis. Hal ini menunjukkan SARIMA efektif digunakan untuk peramalan data penjualan jangka menengah.

C. KESIMPULAN

Praktikum ini menunjukkan bahwa metode ARIMA/SARIMA efektif digunakan untuk memodelkan dan meramalkan data deret waktu yang memiliki tren dan pola musiman. Proses uji stasioneritas, differencing, serta pemilihan parameter model yang tepat sangat berpengaruh terhadap akurasi hasil peramalan. Hasil forecasting pada berbagai dataset membuktikan bahwa model yang dibangun mampu mengikuti pola historis data dan memberikan prediksi yang cukup baik untuk periode ke depan.

D. LAMPIRAN

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**LISTING PRAKTIKUM STATISTIKA
MODUL**